

Permutation Groups

Peter Zeman

Department of Algebra
Faculty of Mathematics and Physics
Charles University



February 16, 2026

Preface

These are lecture notes for the course NMAL432 Permutation Groups, taught at the Faculty of Mathematics and Physics, Charles University.

Contents

Preface	i
1 Three Motivating Examples	1
1.1 Geometry: Finite Subgroups of $O(3)$	1
1.2 Combinatorics: Automorphism Groups of Graphs	1
1.3 Algorithms: Testing Isomorphism of Cubic Graphs	1
2 Basic Notions	2

Chapter 1

Three Motivating Examples

1.1 Geometry: Finite Subgroups of $O(3)$

1.2 Combinatorics: Automorphism Groups of Graphs

1.3 Algorithms: Testing Isomorphism of Cubic Graphs

Chapter 2

Basic Notions